

University Course Advising and Progression Archive for Computing Students

Design Document Specification



UNIVERSITY OF
South Carolina

Computing

Team batduckhamsterdolphinturtle

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System Overview

The system our team is currently creating is a sophisticated tool to simplify the academic planning process for computing students at the University of South Carolina, specifically those in computer science and computer information systems. Spanning an eight-semester plan, the system will provide a structured framework for the students, mapping out their graduation path and updating when courses have been completed. This intuitive interface will help streamline the planning process to ensure all computing students can meet their academic requirements efficiently.

Beyond its purpose for students, our system also benefits advisors. Advisors can use the system to monitor students' academic progress, easily view which students are at risk of delayed graduation, offer guidance, and make recommendations based on their current plan. Parents who have signed the necessary FERPA compliance forms can access the system to stay informed of their child's academic progression, enabling them to provide support as well.

This system will be developed using Java for the back end and JavaFX for the front end, helping our system offer a seamless user experience. Java's performance makes it an ideal choice for the back end, while the capability of JavaFX makes it the perfect option for an interactive front end. Together, these technologies will ensure that this system not only will meet the functional needs of its users but can provide a fresh, engaging experience for academic tracking.

References

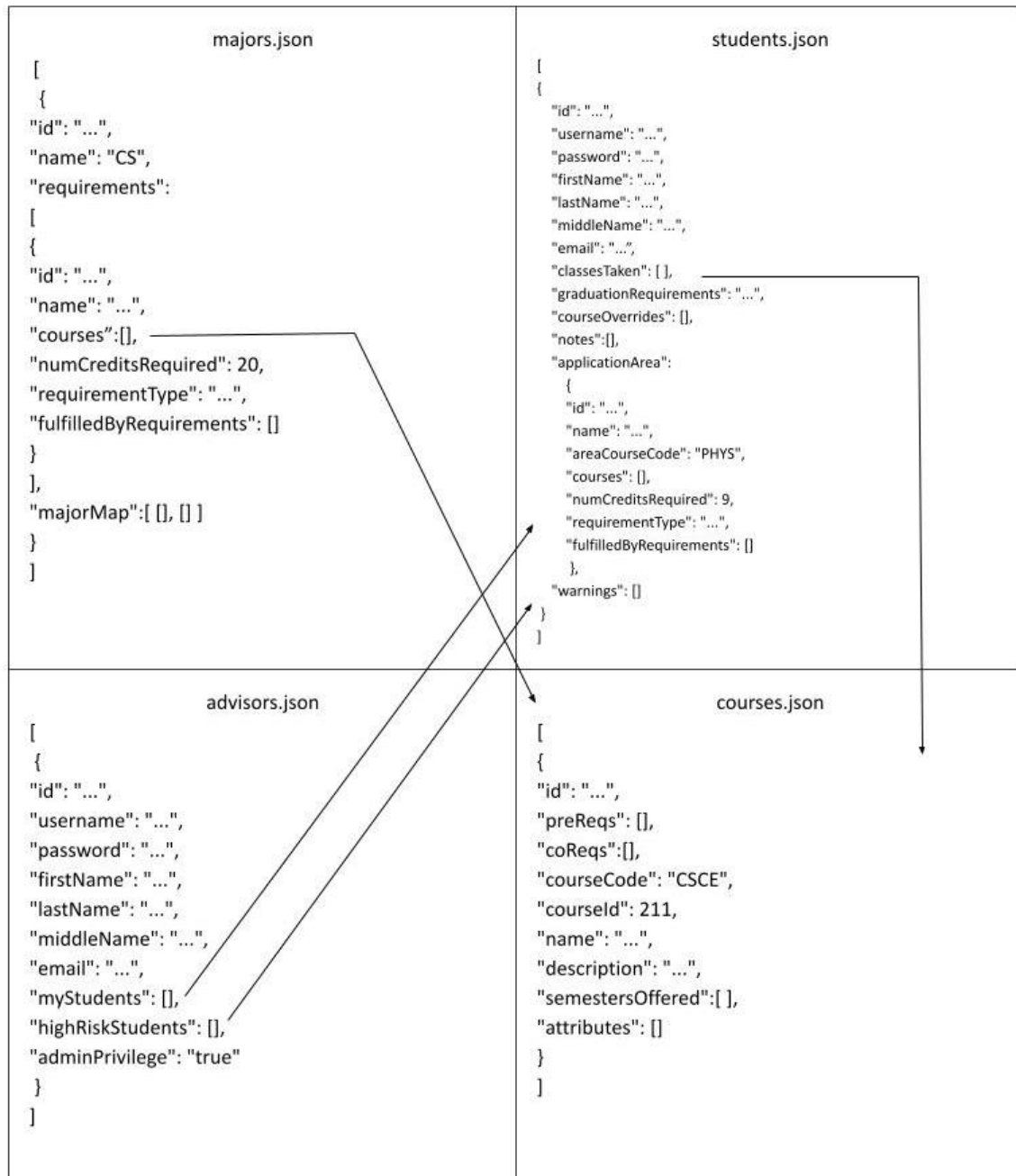
- [Requirements Spreadsheet](#)
- [Requirements Document](#)

Environment Overview

This application is designed to operate on a single machine, leveraging Java to interact with several JSON files that store information about available courses, users, and majors. It facilitates updates to where the user can make changes. In the initial stages, the application will utilize a console or terminal-based interface for both the input and the output. As development progresses, the plan includes transitioning the user interface to a more user-friendly graphical interface with the use of JavaFX.

Data Storage

The data will be stored in four JSON files: `courses.json`, `majors.json`, `students.json`, and `advisors.json`. Each JSON file has objects with a unique identifiers, with the files for users being split up into students and advisors. The diagram below shows how the four JSON files interact. The contents of the variables are shortened for readability in this diagram, but can be viewed in greater detail below.



courses.json:

```
[
  {
    "id": "29aa5281-4f98-4ca8-8d35-b7687b67c1f3",
    "preReqs":
    [
      {
        "faaac848-7dbd-478d-9a18-a73c7df26c17" : "D"
      }
    ],
    "coReqs":[],
    "courseCode": "CSCE",
    "courseId": 211,
    "name": "Digital Logic Design",
    "description": "Number systems, Boolean algebra, logic design, sequential machines.",
    "semestersOffered":[ "Fall", "Spring", "Summer" ],
    "attributes": []
  },
  {
    "id": "4a0cbe0e-3275-46c0-8c62-5be3f9d8fefe",
    "preReqs":
    [
      {
        "242ebdb9-c4f6-4c23-9cbb-573b1e672c4a" : "C"
      }
    ],
    "coReqs":[],
    "courseCode": "CSCE",
    "courseId": 247,
    "name": "Software Engineering",
    "description": "Fundamentals of software design and development; software
implementation strategies; object-oriented design techniques; functional design techniques;
design patterns; design process; source control; testing.",
    "semestersOffered":[ "Fall", "Spring" ],
    "attributes": []
  }
]
```

majors.json:

```
[
  {
    "id": "94ae384a-d123-4914-8109-4bb7f2d346e6",
    "name": "CS",
    "requirements":
    [
      {
        "id": "b6ecf4e6-8ad6-4e62-bc1a-087e4a3f825c",
        "name": "Lower Division Computing",
        "courses": ["29aa5281-4f98-4ca8-8d35-b7687b67c1f3", "d1fdb3a1-1781-4a33-b93f-a658226a11d3" ],
        "numCreditsRequired": 22,
        "requirementType": "Program",
        "fulfilledByRequirements": ["b9579427-bb74-462f-83e1-8f52c0ca91ef", "6183f464-c7db-4352-a90c-4add9824e2ab"]
      },
      {
        "id": "fd3d7002-17ea-4012-96c4-ddef2346929f",
        "name": "Higher Division Computing",
        "courses": ["8e97fbd8-e41a-417c-88a7-7912ce58506e", "a72059fd-4b7e-4ed3-a2bb-0f96e7ae4c8a"],
        "numCreditsRequired": 20,
        "requirementType": "Program",
        "fulfilledByRequirements": []
      }
    ],
    "majorMap":[ ["32b15c31-56a9-40c8-803b-4e933b04419b", "b7c377ed-16e1-4b96-9c9e-910564e432a9"], ["fa99e78b-f5f4-4993-a38d-3f0b168d9a0b", "1205e2a2-3676-4e41-a8f9-92d1f04513d5"] ]
  },
  {
    "id": "6e0da04d-29c3-405b-8320-f33850b3fe86",
    "name": "CIS",
    "requirements":
    [
      {
        "id": "41e56df1-0c85-4190-a917-fabc1363f39e",
        "name": "Foundational Courses",
        "courses": ["c9527972-2282-480c-8f32-2e15441c8ded", "b7847e62-169e-44ed-b140-238c04c1a629"],
        "numCreditsRequired": 12,
        "requirementType": "Program",
        "fulfilledByRequirements": ["a7cef6e1-5c14-469d-9b97-d9ffe7cce7c3"]
      },
      {
        "id": "0d0679da-8990-4988-ac72-2050e4d818bc",
        "name": "Business Information Management Minor",
        "courses": ["ad65b7a1-e96b-47e5-8050-9ef50be3266a", "772050c5-a0cf-4d09-9ea7-02b16f7f75ed"],
        "numCreditsRequired": 18,
        "requirementType": "Program",
        "fulfilledByRequirements": []
      }
    ],
    "majorMap":[["f19dc488-2ac1-4a34-88e2-ae38c243daa3", "40c3a7a8-7300-4dd5-ad7e-d84ef0eb2c80"], ["a1108932-f52c-4dcf-a51d-6be305b95467", "abbe8433-e61e-4d93-bd82-7f0bebe943f1"]]
  }
]
```

students.json:

```
[
  {
    "id": "2d2b31fc-8a28-4793-a5a7-d0cc6ea0f42f",
    "username": "bscileppi",
    "password": "password",
    "firstName": "Blaise",
    "lastName": "Scileppi",
    "middleName": "A",
    "email": "bscileppi@email.sc.edu",
    "classesTaken":
    [
      {
        "course": "fe5abe56-510b-48e9-b10a-a96a1a6d1eb2",
        "yearTaken": "Y2022",
        "semesterTaken": "Fall",
        "grade": "A",
        "creditSource": "Transfer"
      },
      {
        "course": "29aa5281-4f98-4ca8-8d35-b7687b67c1f3",
        "yearTaken": "Y2023",
        "semesterTaken": "Fall",
        "grade": "A",
        "creditSource": "In_Person"
      }
    ],
    "graduationRequirements": "94ae384a-d123-4914-8109-4bb7f2d346e6",
    "courseOverrides":
    [
      {
        "course": "4a0cbe0e-3275-46c0-8c62-5be3f9d8fefe",
        "requirement": "bd7ebb00-0b84-4542-abcc-8431a318b34f"
      },
      {
        "course": "af01cd1f-9e95-41bc-a3d1-30bd16925ea6",
        "requirement": "b6ecf4e6-8ad6-4e62-bc1a-087e4a3f825c"
      }
    ],
    "notes": [ "Take CSCE 240 Fall 2024 semester", "Maintain a 3.0 GPA"],
    "applicationArea":
    {
      "id": "5343a77d-6477-4b68-bd50-06d1f36a2908",
      "name": "Application Area",
      "areaCourseCode": "PHYS",
      "courses": [],
      "numCreditsRequired": 9,
      "requirementType": "Major",
      "fulfilledByRequirements": []
    },
    "warnings":
    [
      {
        "name": "Honors Credits Warning",
        "description": "Please take at least 45 credits in the SCHC"
      }
    ]
  },
]
```

```

    "id": "4d376e14-de1e-4f54-b975-69dc69d60601",
    "username": "sremaili",
    "password": "password2",
    "firstName": "Soraya",
    "lastName": "Remaili",
    "middleName": "Sakina",
    "email": "sremaili@email.sc.edu",
    "classesTaken":
    [
      {
        "course": "29aa5281-4f98-4ca8-8d35-b7687b67c1f3",
        "yearTaken": "Y2023",
        "semesterTaken": "Spring",
        "grade": "A",
        "creditSource": "Transfer"
      },
      {
        "course": "4a0cbe0e-3275-46c0-8c62-5be3f9d8fefe",
        "yearTaken": "Y2023",
        "semesterTaken": "Fall",
        "grade": "A",
        "creditSource": "In_Person"
      }
    ],
    "graduationRequirements": "94ae384a-d123-4914-8109-4bb7f2d346e6",
    "courseOverrides":
    [
      {
        "courseTaken": "29aa5281-4f98-4ca8-8d35-b7687b67c1f3",
        "requirement": "b6ecf4e6-8ad6-4e62-bc1a-087e4a3f825c"
      }
    ],
    "notes": [ "Remember to fill out the study abroad application", "Plan to take an Honors elective next semester"],
    "applicationArea":
    {
      "id": "5b8c3b9a-9230-47d9-b317-45d3f7940aec",
      "name": "Application Area",
      "areaCourseCode": "MATH",
      "courses": [],
      "numCreditsRequired": 9,
      "requirementType": "Major",
      "fulfilledByRequirements": []
    },
    "warnings":
    [
      {
        "name": "Residence Credits Warning",
        "description": "Please take at least 15 credits in residence"
      }
    ]
  }
]

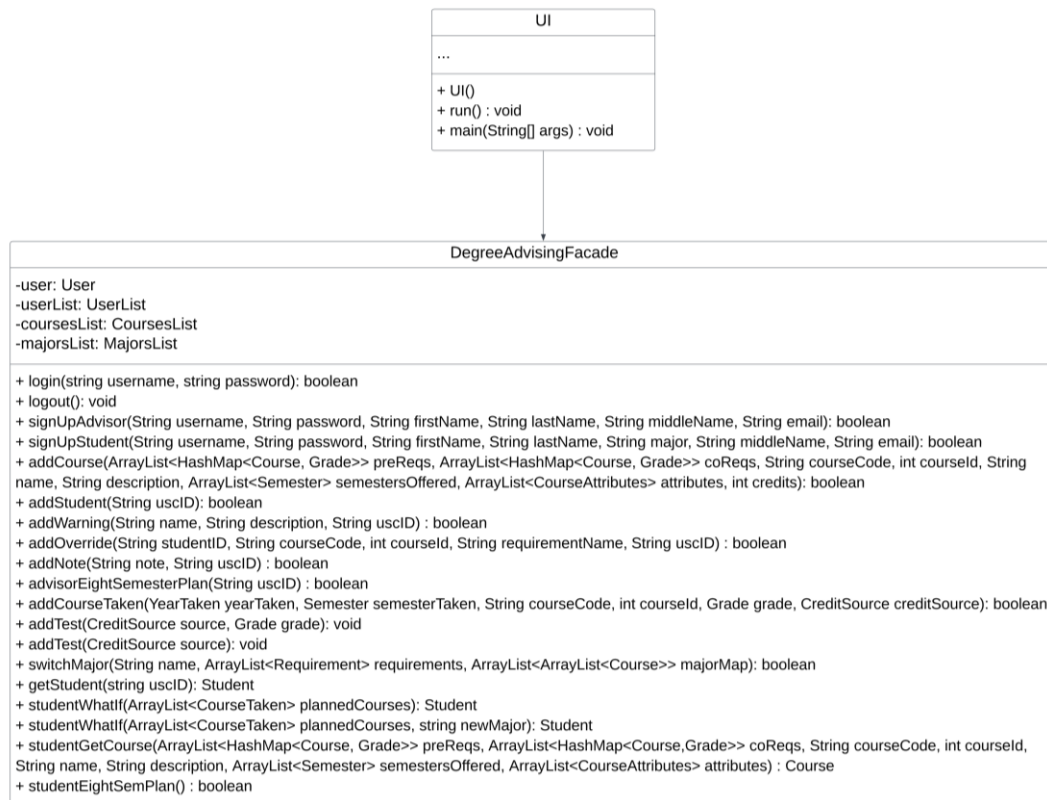
```


advisors.json:

```
[
  {
    "id": "e757bc03-f52a-4cc2-b538-39a645e5c93b",
    "username": "cturner",
    "password": "password3",
    "firstName": "Carrie",
    "lastName": "Turner",
    "middleName": "Winona",
    "email": "cturner@email.sc.edu",
    "myStudents": ["4d376e14-de1e-4f54-b975-69dc69d60601", "2d2b31fc-8a28-4793-a5a7-d0cc6ea0f42f" ],
    "highRiskStudents": ["2d2b31fc-8a28-4793-a5a7-d0cc6ea0f42f"],
    "adminPrivilege": "true"
  },
  {
    "id": "cad42725-d132-48c5-b598-f7e89d75c404",
    "username": "llimbaugh",
    "password": "password5",
    "firstName": "Leah",
    "lastName": "Limbaugh",
    "middleName": "Rachel",
    "email": "llimbaugh@email.sc.edu",
    "myStudents": [ "01c34608-5b3a-456f-97c0-6abe3d0c4d6a", "e2eb1601-35d6-4635-853e-9a959da9cbb3" ],
    "highRiskStudents": ["e2eb1601-35d6-4635-853e-9a959da9cbb3"],
    "adminPrivilege": "false"
  }
]
```

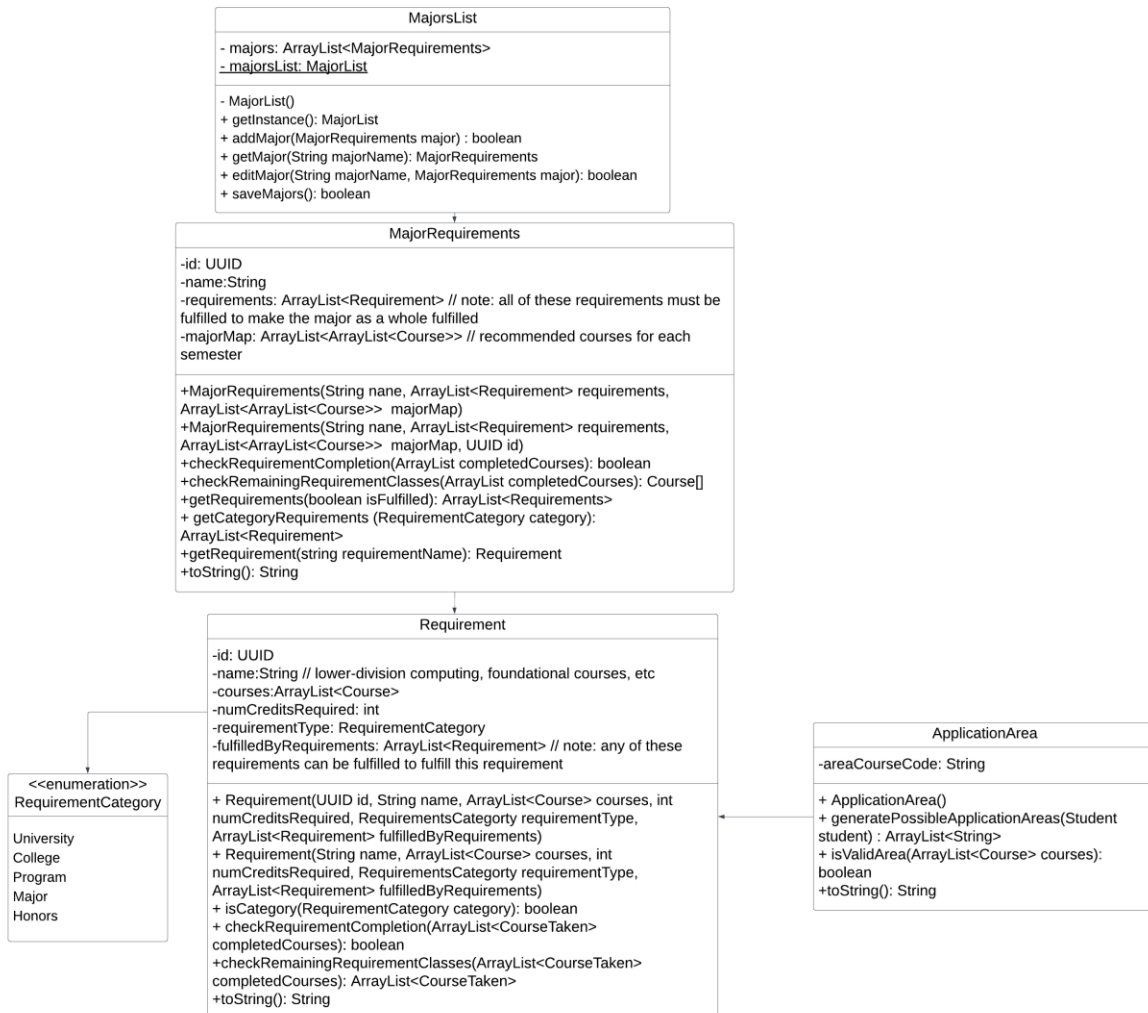
Class Diagram – Structural Design

The UI and Facade



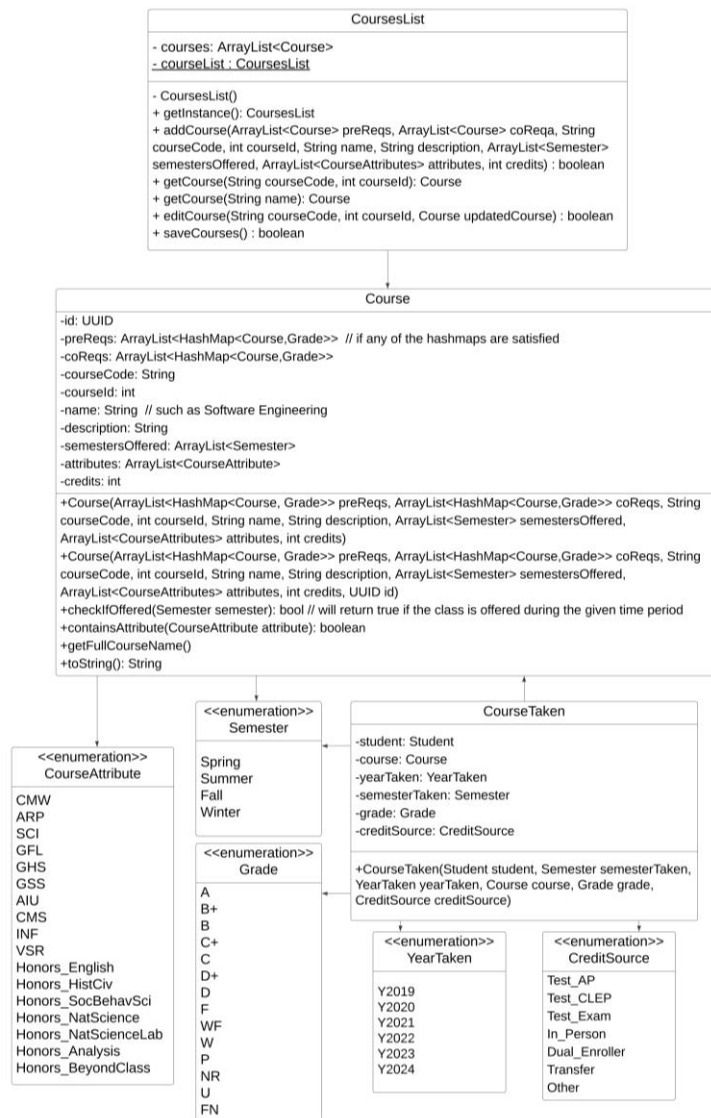
This part of the application is what controls and responds to the user's input. The methods in the facade, which detail the different actions that a user could take, interact with the classes and feed the responses to these actions back into the UI, where the user can see the results of what they did. The facade contains methods for both the student user and the advisor user and enables a user to login and logout of the system while also creating an account within the system. In order to properly pass the methods through the lower classes, the facade must interact with the `UserList`, `CourseList`, and `MajorsList`.

The Requirements



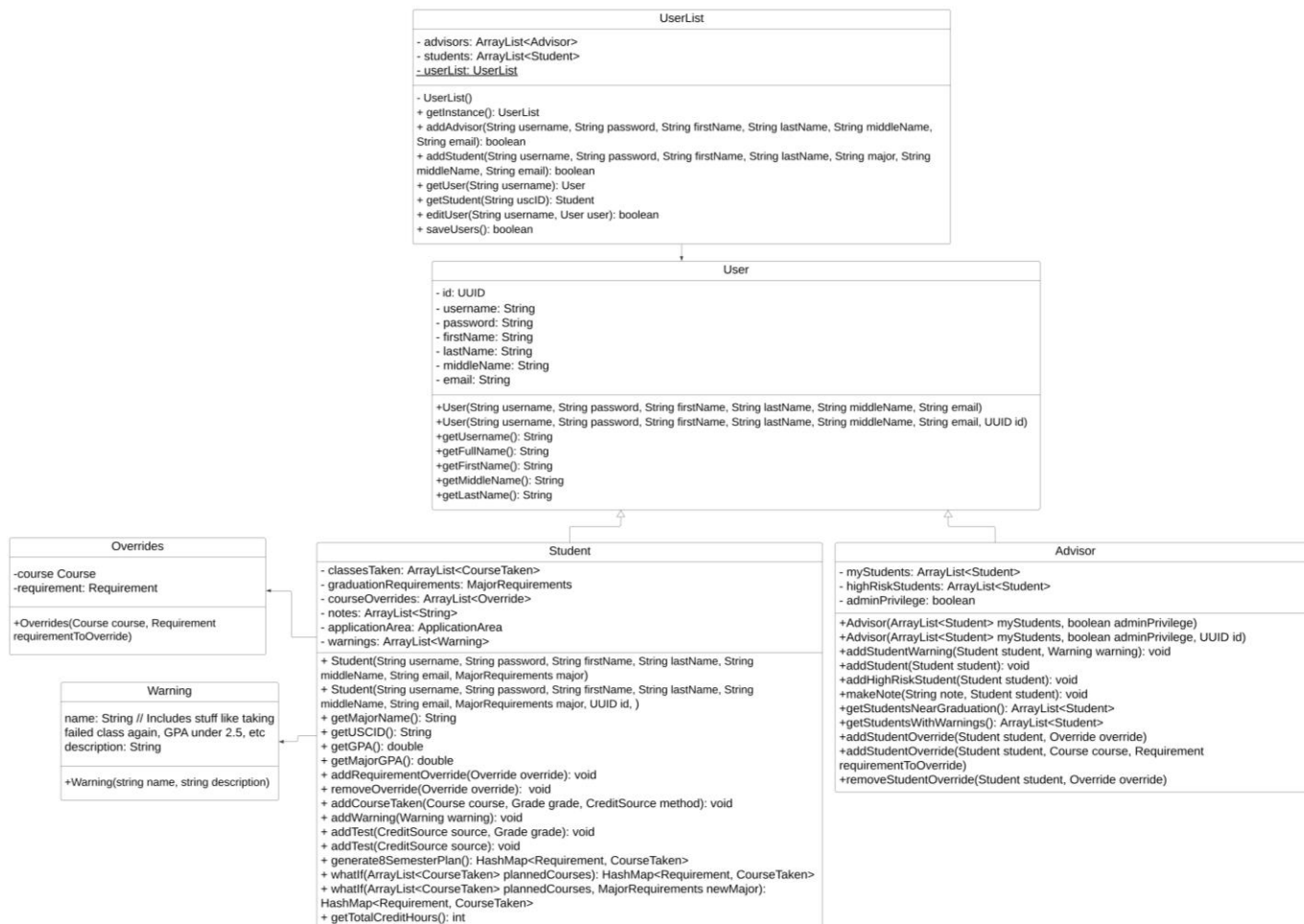
The MajorRequirements drive the logic behind the completion of a student's degree and the making of the What Ifs and Eight Semester Plans. Interacting with the UI and Facade through the MajorsList, the MajorRequirements use Requirements, which can be grouped into certain sections, like 'Lower Division Computing' or 'Foundational Courses'. Leaving it more generic and open ended like this allows for greater mutability if the application were to be extended to incorporate different majors.

The Courses



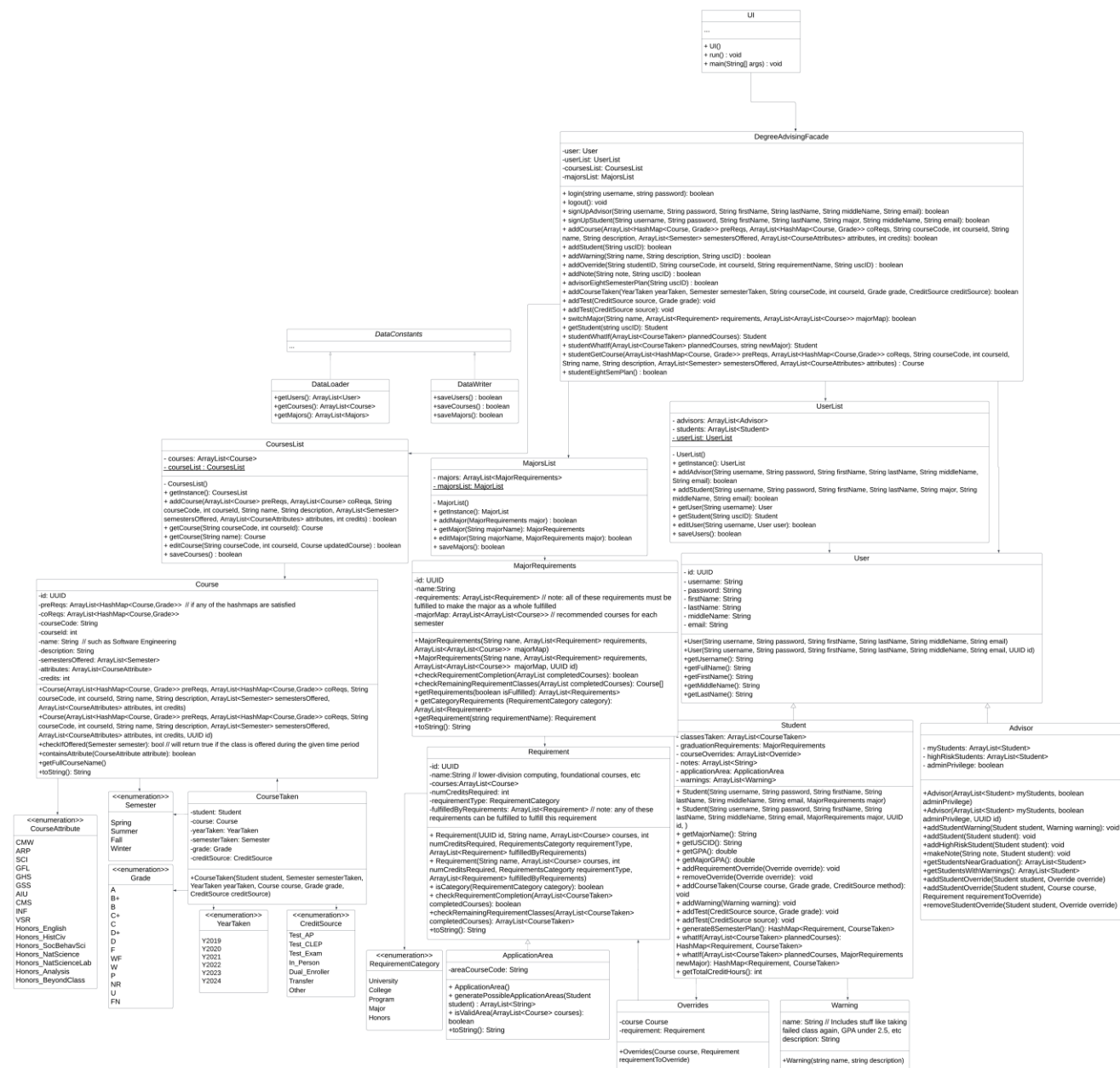
This part of the application controls one of the most crucial parts of a degree advising system, the courses that the degree is made up of. Once again linked to the UI and Facade by a Courses List, the Course application is separated into Course and Course Taken, with CourseTaken having more fields related to when the class was taken. The Course application also makes extensive use of enumerations to detail course attributes, semesters, years, grades, and the source of any given credit.

The Users



This part of the application controls the different types of users that might use the system, in this case, the student and the advisor. Both the Student and the Advisor extend the User class. The Student class also makes use of the Override and Warning classes, so that Students and Advisors can fix system mistakes and the system can warn students when they are not on the right track or fulfilling the correct requirements. Students also have the option to generate What If plans and Eight Semester plans, making their reach of actions slightly broader than that of an Advisor. The Advisor, in contrast, has lists of Students as their private fields and an option to have administrative privileges.

Complete Overview of UML Diagram



Link for increased visibility: [Lucid chart link](#)

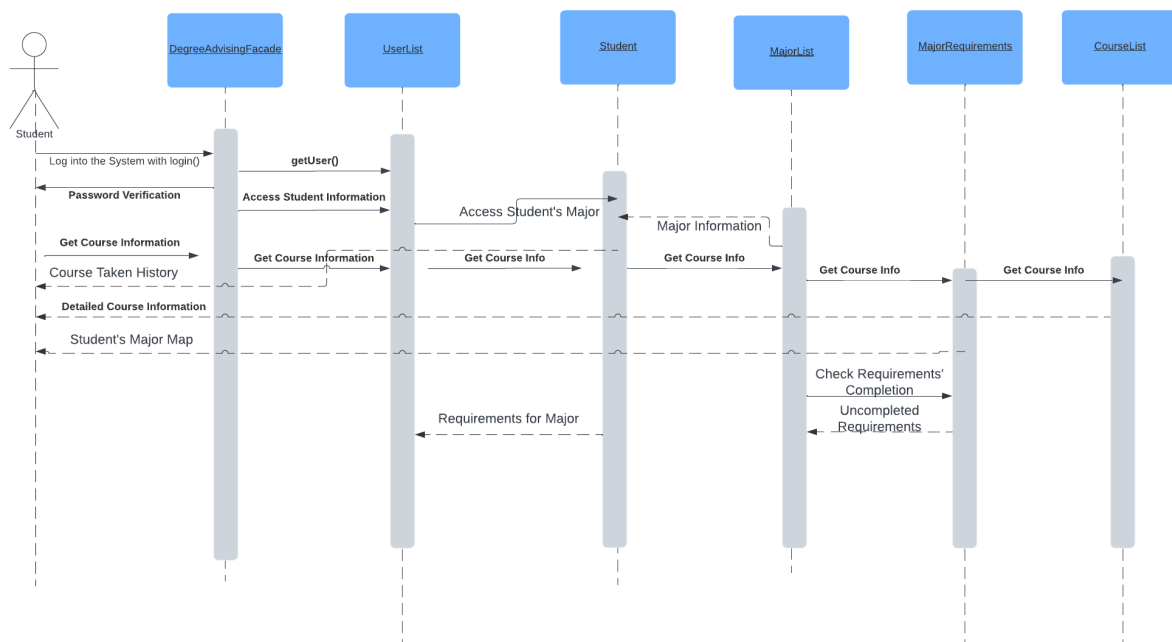
Sequence Diagrams - Dynamic Model

Scenario #1:

Scenario Name: Student Interaction with Completed and Uncompleted Courses

Scenario Description: The student will log into the system, where they will be presented with their major map that lists all of the classes they have taken and will take. While examining the major map, the major map displays that the class, CSCE 247, has not been taken. The student is then able to select and see all the information about the class, like the course description and prerequisites.

Sequence Diagram:



Scenario #2:

Scenario Name: Advisor Interaction with Advisee List and Advisee Warnings

Scenario Description: The advisor logs into the system with their account. The advisor searches for a list of the students whom they advise. The advisor selects a specific advisee to see the students' 8 semester plan. While examining the 8 semester plan, the advisor is given a warning about how the student is at risk of losing their scholarship. The advisor then makes a note on the advisee's account to warn them of the risk.

Sequence Diagram:

